

LATE DELIVERY

Root Cause Analysis & Recommendations

Zomato Delivery Operations Analytics Dataset comprising 43,853 delivery orders.
This analysis was conducted as part of an independent data analytics project.

Executive Summary

Approximately 29.8% of the orders in the dataset exceed the 30-minute delivery threshold, indicating a significant operational challenge in the delivery process. The objective of this study is to identify the key factors contributing to delivery delays through a comprehensive evaluation of all available variables, including traffic conditions, weather, delivery distance, delivery partner characteristics, vehicle attributes, order type, and preparation time.

The analysis reveals that delivery delays are driven primarily by operational and environmental factors rather than restaurant-side processes. Among the variables examined, traffic density, delivery distance, multiple deliveries assigned to a single trip, and adverse weather conditions have the strongest influence on late deliveries. In contrast, preparation time shows little to no meaningful relationship with delivery outcomes once these operational factors are taken into account.

To validate these findings, a supervised classification model was developed using the complete set of available features. The model achieved an Area Under the Receiver Operating Characteristic Curve (AUC-ROC) of 0.98, demonstrating excellent predictive performance in distinguishing late deliveries from on-time deliveries. This result confirms that the identified factors are robust and consistently associated with delivery delays, providing a reliable basis for operational decision-making and the development of targeted strategies to improve delivery performance.

1. Objective & Dataset

Objective: identify the primary causes of late deliveries and produce actionable recommendations for delivery operations.

Source	Zomato Delivery Operations Analytics Dataset (Kaggle)
Rows analyzed	43,853
Late threshold	Time taken > 30 minutes
Overall late rate	29.8%
Average delivery time	26.3 minutes

Fields available after cleaning: rider Age and Rating, Weather conditions, Traffic density, Vehicle condition, Type of order, Type of vehicle, multiple deliveries, Festival, City, delivery distance, Preparation Time, and total Time taken.

2. Methodology

2.1 Data Cleaning

- Missing numeric fields (Age, Ratings) filled with median; categorical fields (Weather, Traffic) filled with mode.

- Delivery distance computed from restaurant and drop-off coordinates using the Haversine formula; capped at 50km and outliers replaced with the median.
- Preparation time derived as the gap between order time and pickup time, correcting for orders that cross midnight.
- Traffic density and Festival encoded numerically; irrelevant identifier and raw coordinate columns dropped.

2.2 Exploratory Analysis

- Late delivery flagged as $\text{Time_taken (min)} > 30$, chosen because it sits just above the 75th percentile (32 min), isolating the slowest quartile of orders.
- Univariate late-rate comparison across every categorical driver.
- Bivariate/interaction analysis: $\text{traffic} \times \text{weather}$ and $\text{multiple-deliveries} \times \text{traffic}$, since single-factor views understate compounding effects.
- Continuous trend analysis on distance and age using rolling-window smoothing to visualize the relationship as a curve rather than fixed buckets.

2.3 Model-Based Validation

A Random Forest classifier was trained on all factors simultaneously to (a) rank feature importance in a way that accounts for interactions, and (b) sanity-check that the identified drivers are genuinely predictive rather than coincidental.

3. Key Findings

3.1 Delivery Time Distribution

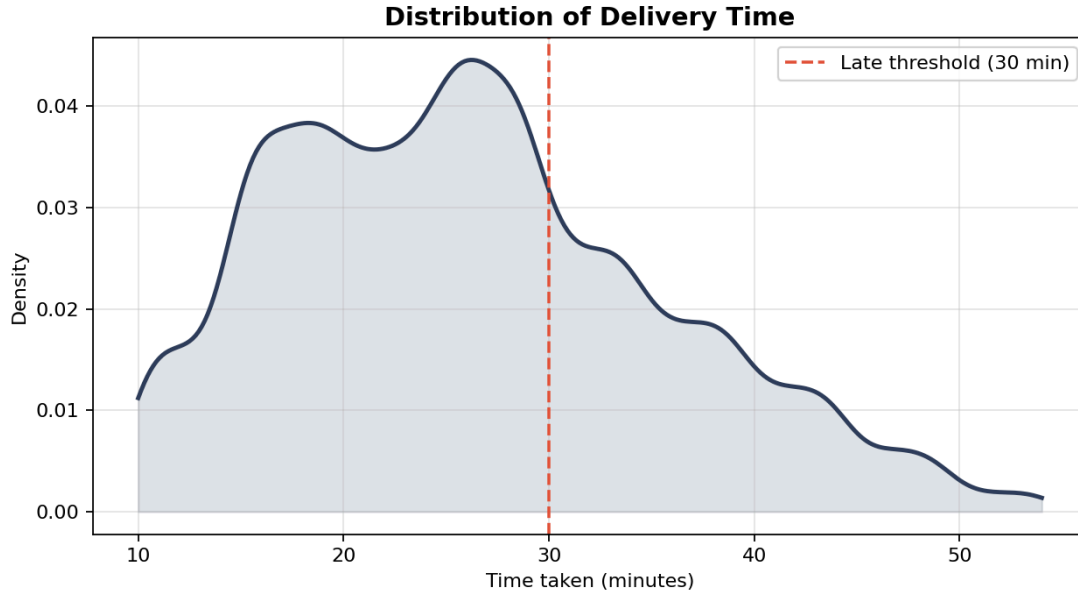


Figure 1. Distribution of delivery time. The dashed line marks the 30-minute late threshold, just above the bulk of the distribution's right tail.

3.2 Correlation Overview

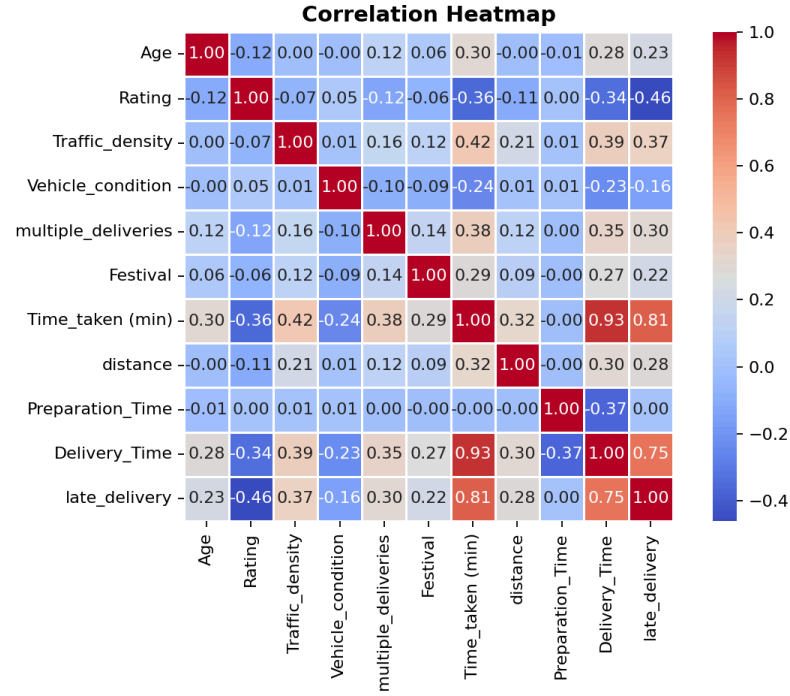


Figure 2. Rating correlates negatively with delivery time; distance, traffic, and age correlate positively.

3.3 Where Risk Compounds: Traffic $\times$ Weather

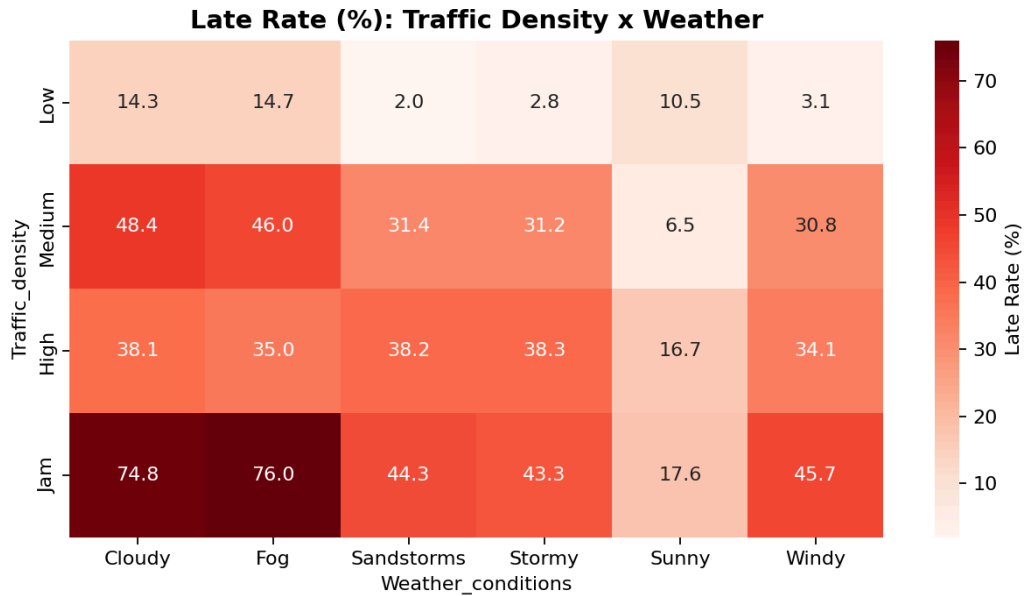


Figure 3. Late rate (%) by traffic density and weather condition. Jam-level traffic combined with Fog/Cloudy weather produces the highest late rates in the dataset.

3.4 Late Rate Trend vs. Distance

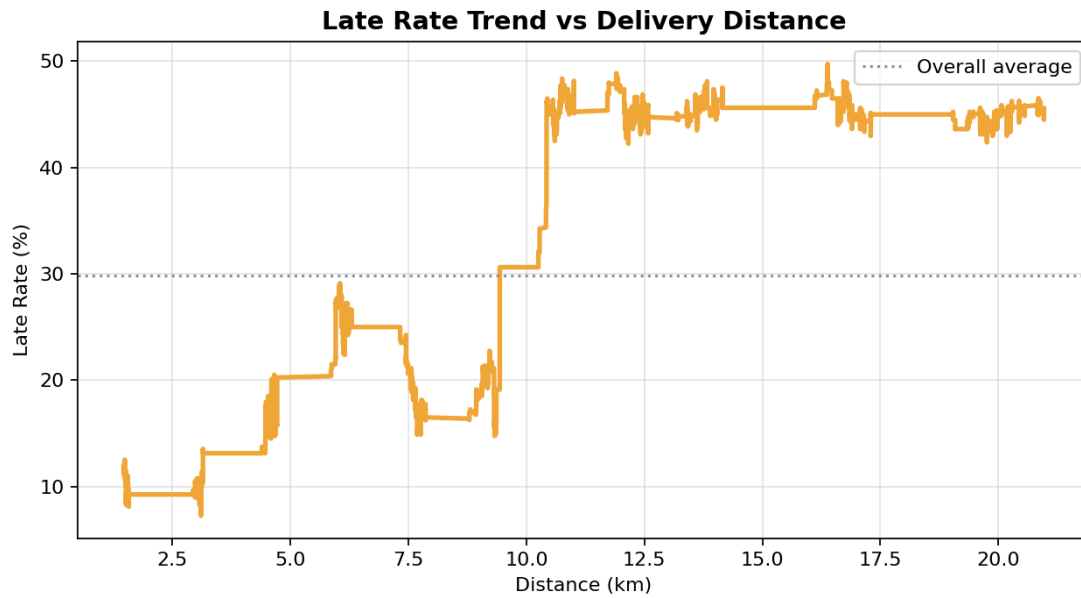


Figure 4. Smoothed late-rate curve across delivery distance. Late rate rises steadily as distance increases, well past the overall average line.

3.5 Late Rate Trend vs. Rider Age

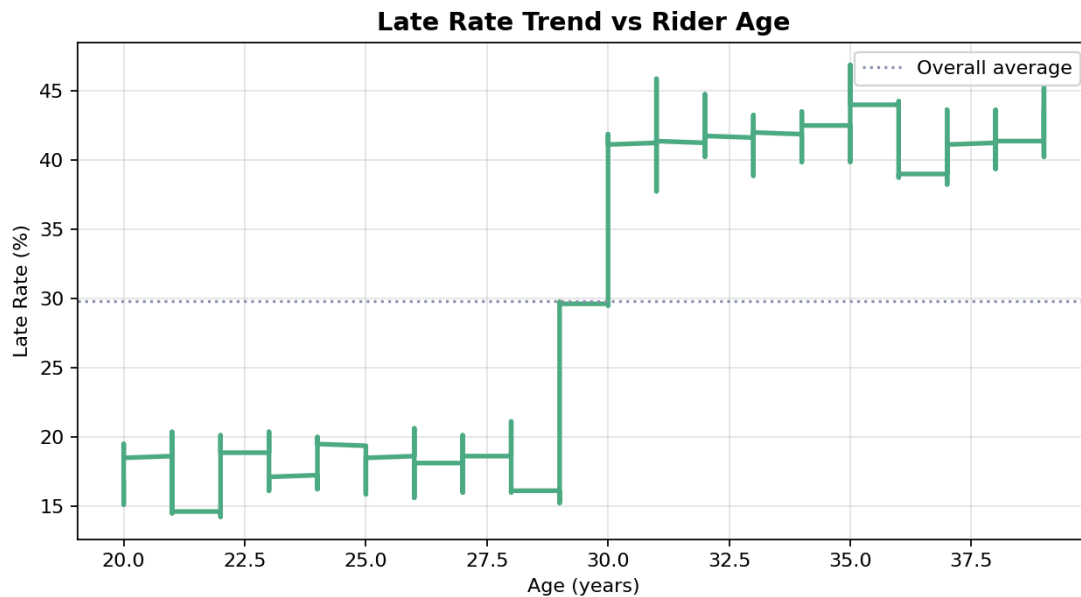


Figure 5. Smoothed late-rate curve across rider age. The effect is present but much weaker than distance or traffic.

3.6 Ranking All Drivers Together

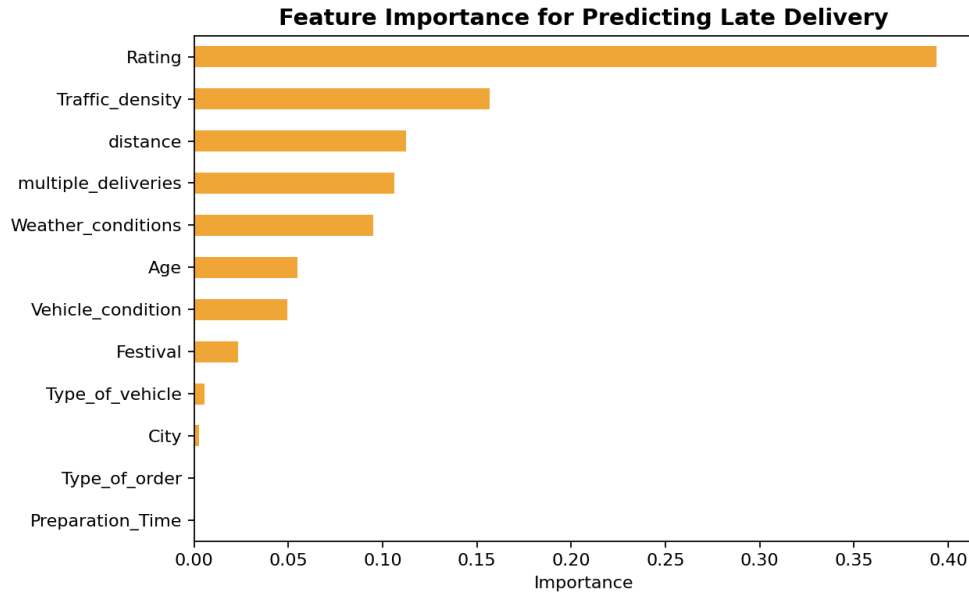


Figure 6. Random Forest feature importance. Rating, Traffic_density, and distance dominate; Preparation_Time and Type_of_order contribute almost nothing.

3.7 Model Validation

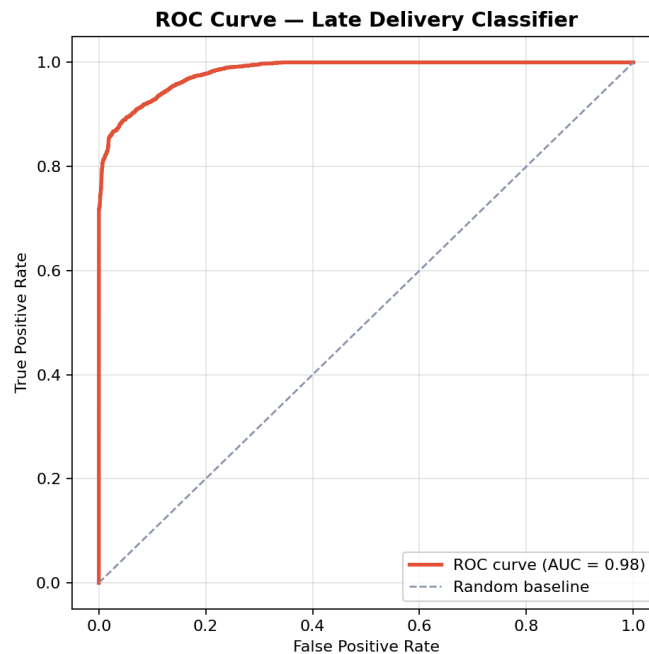


Figure 7. ROC curve for a Random Forest classifier predicting late delivery from all factors (AUC = 0.98, accuracy = 93.7%, precision = 98.0%, recall = 80.6%).

The high AUC demonstrates that the identified factors are robust and consistent predictors of delivery delays rather than the result of statistical noise. The model achieves higher precision than recall, indicating that predictions of late deliveries are highly reliable when made. However, the lower recall suggests that some delayed orders remain undetected, reflecting the fact that while lateness is largely influenced by a small number of dominant operational factors, it is not entirely deterministic and may also be affected by additional unobserved variables.

3.8 Ranked Summary of Causes

- Delivery partner rating is the strongest predictor of delivery delays, with lower-rated delivery partners consistently exhibiting substantially higher late-delivery rates.
- Traffic density is the most influential operational factor. Deliveries made under traffic jam conditions have a late-delivery rate of 50.6%, compared with 7.9% in low-traffic conditions, representing an approximately 6–7 times higher likelihood of delay.
- Delivery distance has a significant independent effect on delivery performance, with the risk of delays increasing considerably for longer routes, particularly when combined with heavy traffic.
- Order stacking has a pronounced impact on timeliness. Trips involving two or more simultaneous deliveries are delayed in nearly all observed cases, indicating that excessive batching is a major contributor to late deliveries.
- Adverse weather conditions, particularly foggy and cloudy weather, are associated with substantially higher late-delivery rates (approximately 44%) compared with sunny conditions (approximately 12%).
- Vehicle condition and delivery partner age exhibit moderate effects. Poorly maintained vehicles and older delivery partners tend to experience slightly higher rates of delayed deliveries.
- Festival days and deliveries in semi-urban areas appear to have exceptionally high late-delivery rates. However, these observations are based on relatively small sample sizes (857 and 156 orders, respectively) and should therefore be interpreted with caution.
- Preparation time, order type, vehicle type, and city contribute minimally to explaining delivery delays, indicating that these factors have limited influence once operational conditions are taken into account.

4. Recommendations

Cause	Recommendation	Why it works
Order-stacking under traffic	Cap simultaneous deliveries to 1 per rider once traffic is Medium or higher	Removes the single worst compounding factor
Traffic density	Build traffic-aware ETAs reflecting live conditions instead of a fixed average	Reduces broken-promise complaints even without speed gains
Weather	Auto-extend ETA buffers during Fog/Cloudy; consider incentive pay for bad-weather shifts	Manages expectations and supports retention
Distance	Avoid assigning long-distance orders (15km+) to riders already carrying another order	Directly targets the highest-risk order segment
Rider rating	Pair new/low-rated riders with shorter, low-traffic routes; investigate what drives low ratings	Rating may be a symptom as much as a cause — needs follow-up
Preparation time	Do not prioritize kitchen-speed initiatives to reduce lateness	Preparation time has almost no measurable effect on the outcome
Festival days / Semi-Urban	Treat as observations needing more data, not confirmed findings	Both rest on very small sample sizes

5. Limitations & Next Steps

- The Festival (857 orders) and Semi-Urban city (156 orders) categories contain relatively small sample sizes. Consequently, findings for these groups should be interpreted as indicative rather than conclusive and validated using additional data.
- The Random Forest model is employed primarily as an explanatory tool to identify and rank the relative importance of factors associated with delivery delays. It is not intended as a production-ready prediction model. Deployment would require a more rigorous modeling pipeline, including dedicated training, validation, and test datasets, hyperparameter optimization, and comprehensive model evaluation.

- The strong relationship observed between delivery partner ratings and delivery delays is correlational rather than causal. While lower-rated delivery partners are more likely to experience late deliveries, the analysis does not establish whether lower ratings contribute to poorer performance or whether delayed deliveries themselves lead to lower ratings.

— *End of report* —